

DR. LIAM MOROY

Research Associate in Deep Generative Modelling for Inverse Problems.

✉ l.moroy@hw.ac.uk

📍 Edinburgh, Scotland, UK

🔗 <https://liammry.github.io/>



SUMMARY

Research Associate in the BISC group at Heriot-Watt University, under the supervision of Yoann Altmann and Steve McLaughlin. My research addresses the resolution of highly ill-posed inverse problems in physical sciences, leveraging deep generative models, with a focus on diffusion-based approaches, to achieve robust and uncertainty-aware reconstructions.

EDUCATION

Ph.D. – ONERA, DTIS/MIC group

- Université Paris-Saclay.
- Generative AI (diffusion models) for solving inverse problems in imaging, applied to tomographic imaging.
- Supervised by Prof. Jean-François Giovannelli, Dr. Frédéric Champagnat, and Dr. Guillaume Bourmaud.

📅 2022 - 2025

📍 Paris, France

MSc – Complex Systems Engineering

- Concurrent with the below master degree.
- University of Bordeaux.

📅 2022

📍 Bordeaux, France

MSc - Electronics Engineering, specialized in Signal Processing

- Enseirb-Matmeca engineering school, Bordeaux INP.

📅 2019 - 2022

📍 Bordeaux, France

BSc - Electronics Engineering

- Enseirb-Matmeca engineering school, Bordeaux INP.

📅 2017 - 2020

📍 Bordeaux, France

RESEARCH ACTIVITIES

Publications:

- “On the Posterior Gap in Plug&Play Diffusion Methods for Sparse-View CT”, IEEE JSTSP 2026 Special Issue on High-Dimensional Imaging: Emerging Challenges and Advances in Reconstruction and Restoration, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, ICASSP 2025, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.

Oral Presentations:

- “Improving the Posterior Sampling of Plug & Play Diffusion Models by Super-Calibration : application to Poisson inverse problem” 7th SLSIP Workshop, May 2026, Krujë, Albania.
- “Plug & Play Diffusion models for general imaging inverse problems” ISSS Seminar, Mar 2026, Heriot-Watt University, Edinburgh, UK.
- “Plug-and-Play Diffusion Methods for Posterior Sampling in Tomographic Reconstruction”, AI in IMS, Jan. 2025, Bordeaux.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, GdR IASIS: Advances in Learning-Based Image Restoration, Dec 2024, Henri Poincaré Institute, Paris.

ACADEMIC EXPERIENCE

University of Bordeaux

- Master ISC – MSc teaching assistant - Introduction to noise and random signals (2023/24 & 2024/25).

ENSEIRB-MATMECA engineering school

- Engineering cycle – MSc teaching assistant - Generative modelling with diffusion models (2024/25).
- Engineering cycle – Freshman teaching assistant - Introduction to programming in C/C++ (2023/24 & 2024/25).

PROFESSIONAL EXPERIENCE

Research Associate – Generative AI for Inverse Problems in Physical Sciences

Heriot-Watt University (BISC)

📅 Dec. 2025 - Present

📍 Edinburgh, Scotland

- Development and application of deep generative models (diffusion models) as learned priors for solving highly ill-posed inverse problems across multiple physical imaging modalities: Computed Tomography for nuclear applications, non-conventional optics, and medical imaging.

Python Pytorch/CUDA Generative AI Bayesian Inversion

R&D Internship – Deep Learning for tomographic reconstruction of compressible flows

ONERA (DTIS/MIC)

📅 Feb. 2022 - Aug. 2022

📍 Paris, France

- Design and evaluation of learned prior models for Bayesian CT reconstruction of compressible flows from limited measurements.

Python Pytorch/CUDA Generative AI Bayesian Inversion

R&D Internship – Deep Learning for nebula removal in astronomical images

Astrophysics Laboratory of Bordeaux (LAB) & IMS Laboratory

📅 Jul. 2021 - Sep. 2021

📍 Bordeaux, France

- Training a regressor from noisy ground truths and transfer learning from synthetic datasets.

Python Pytorch/CUDA C/C++ Image Processing

Internship – FPGA implementation of an encryption algorithm

IMS Laboratory (Circuit Design/CSN)

📅 Jul. 2020 - Aug. 2020

📍 Bordeaux, France

- Implementation of an AES (Advanced Encryption Standard) algorithm for a RISC-V processor on a Zynq 7020.

Verilog VHDL C/C++ Cryptography

SKILLS

Programming

- Python, C, C++

ML & Scientific Libraries

- PyTorch / CUDA
- NumPy, Matplotlib

Tools & Workflow

- Git / GitHub
- $\LaTeX$
- Linux / Bash
- Slurm / HPC clusters

Research Expertise

- Diffusion / score-based generative models
- Plug-&-Play methods
- Bayesian inference & inverse problems
- Signal & image processing

Languages

- French (native)
- English (TOEIC 950 – fluent)
- Spanish (B1 – intermediate)
- Japanese (beginner)